ASSIGNMENT 1

Q. Given a dataset with repetitive patterns, design a simple physical model for lossless compression. Justify how model reduces data size and provide an example to demonstrate your approach.

To design a simple physical model for lossless compression on datasets with repetitive patterns, we can use Run-Length Encoding (RLE). This is one of the simplest and most effective compression techniques when data contains repeated values.

CONCEPT :-

*- Instead of storing repeated data values individually, RLE stores ;

→ value

→ the number of times it repeats

*- Each sequence is encoded as :

(value, count)

→ COMPRESSION ALGORITHM :-

- Encoding

1. Read the dataset sequentially.
2. Count consecutive occurences of the same value.
3. Replace repeated values with a pair.
4. Continue until the end of the dataset.

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- Decoding

1. Read each (value, count) pair.
2. Reconstruct the original data by repeating the value 'count' times.

→ JUSTIFICATION :-

* Let :

  Original size = no. of elements N

  Compressed size = no. of runs R

* If data contains many repetitions :

  $R << N$

  ie, Compression Ratio = $N / 2R$

Eg :- Consider dataset AAAABBCCCCCAA

Step 1 - Identify runs

- A repeated 4 times
- B repeated 2 times
- C repeated 5 times
- A repeated 2 times

Step 2 - Encode

(A,4), (B,2), (C,5), (A,2)

Step 3 - Size comparison

- Original length = 13 elements

  Compressed length = 4 pairs (8 values)

ie, Compression Achieved

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IMPLEMENTATION :-

```
def rle_encode (data) :
    encoded = []
    count = 1
    for i in range (1, len(data)) :
        if data[i] == data[i-1] :
            count += 1
        else :
            encoded.append ((data[i-1], count))
            count = 1
    encoded.append ((data[i-1], count))
    return encoded

def rle_decode (encoded) :
    decoded = []
    for (value, count in encoded ) :
        decoded.extend ( [value] * count )
    return decoded

data = ["A", "A", "A", "A", "B", "B", "C", "C", "C", "C", "C", "A", "A"]

encoded = rle_encode (data)
decoded = rle_decode ( encoded )

print (" Original : ", data )
print (" Encoded : ", encoded )
print (" Decoded : ", decoded )
```

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→ ADVANTAGES :

* - Simple to implement
* - Lossless, that is, original data is fully recoverable.
* - Effective for repetitive datasets.

LIMITATIONS :

* - Not efficient for non-repetitive data.
* - May increase size if data has no repetition.

The RLE model compresses data by replacing repeated sequences with (value, count) pairs.

It reduces data size by eliminating redundancy, making it highly effective for datasets with repetitive patterns while ensuring lossless reconstruction.

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